AI Elites at NADI 2026 Shared Task: Selecting on Out-of-Domain Accuracy for Cross-Domain Spoken Arabic Dialect Identification

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Abstract

We describe our submission to NADI-2026 Subtask 2, country-level spoken Arabic dialect identification on a blind test set from undisclosed sources. Because the target domain is unknown at development time, we select on out-of-domain accuracy rather than in-domain validation accuracy at every decision point, comparing three backbones under a common pooling-plus-linear head and training the selected Arabic ASR encoder on ADI20-micro mixed with Casablanca. The system reaches 91.54% in-domain, 73.10% on the held-out MADIS-5 corpus (scored five-way), and 52.51% on the blind test set (C_{avg} 0.11), placing fourth. The criterion helped, choosing an encoder that in-domain accuracy would have rejected by 32 points on a held-out corpus, a gap more than twenty times our measured seed variation, but it was not sufficient: of the 39.03-point in-domain to blind gap, at least 18.44 points are attributable to leaving the training domain. We trace part of the remainder to a corpus that entered the training mixture while still being used for checkpoint selection.

1 Introduction

Spoken Arabic varies substantially across the roughly twenty Arabic-speaking countries, yet most speech technology is trained on Modern Standard Arabic or a narrow subset of dialects. Automatic dialect identification (ADI) is a useful testbed for acoustic domain generalization: dialect labels are entangled with recording conditions, channel characteristics, and speaker identity, so a model can post high in-domain accuracy without having learned dialectal signal at all.

NADI has run annually since 2020, for five editions as a text-based benchmark (Abdul-Mageed et al., 2020, 2021, 2022, 2023, 2024), moving to speech in 2025 (Talafha et al., 2025). NADI-2026 Subtask 2 (Sullivan et al., 2026) makes the cross-domain failure mode explicit: given a raw utterance, a system predicts its dialect from a fixed inventory of twenty classes (Elleuch et al., 2025a), scored by top-1 accuracy on a blind test set whose sources and recording conditions differ from the training data, with C_{avg}, a NIST LRE-style cost, secondary. The label space is closed and admits no abstention, so neighbouring Levantine, Gulf, or Maghrebi varieties may be near-indistinguishable and some error is irreducible; the blind set moreover covers an undisclosed subset while submissions must score over all twenty classes, exposing systems to false positives on absent ones.

Prior ADI work is strong but tuned in-domain. In NADI-2025 the winning ELYADATA system fine-tuned a Whisper-large-v3 encoder (Radford et al., 2023) with attentive pooling to 79.8% accuracy (Elleuch et al., 2025b), while refinement of the ECAPA-TDNN baseline (Desplanques et al., 2020) yielded little headroom (Rafiul Biswas et al., 2025).

That domain shift breaks such systems is established rather than speculative, and not only for Arabic. Sullivan et al. (2023) measure the vulnerability of Arabic ADI tools to out-of-domain data directly and find self-training an insufficient remedy under realistic conditions, while Shon et al. (2019) treat an unknown target domain as the design constraint rather than an afterthought. The magnitude is comparable in neighbouring tasks: in Slavic spoken language identification, in-domain accuracy above 95% falls to between 37.8% and 54.0% across a read-speech to radio-broadcast shift, a drop of 43 to 61 points (Abdullah et al., 2020). Proposed remedies converge on suppressing the nuisance factor rather than modelling it: domain-adversarial training through a gradient reversal layer (Ganin et al., 2016), applied to accented ASR by Sun et al. (2018) and to cross-corpora LID by Dey et al. (2023), who pair it with augmentation-based domain diversification and report that the domain-invariant variant is what improves cross-corpora error while the domain-aware variant helps same-corpus per-

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 88–94
October 28-29, 2026 ©2026 Association for Computational Linguistics

formance instead; an auxiliary loss that discards channel-specific content without requiring channel labels (Muralikrishna and Dinesh, 2022); and, for Arabic ADI specifically, speaker normalisation through voice conversion (Abdullah et al., 2025b), decisive in the NADI-2025 cross-domain setting (Abdullah et al., 2025a).

What this literature does not supply is an account of how well any development-time proxy predicts blind cross-domain performance. Each method above is validated against a shift its authors could measure. Because the target domain here is unknown at development time, any out-of-domain corpus used for selection is a guess about the shift rather than a measurement of it, and whether such a guess transfers is an open empirical question.

We therefore treat cross-domain behaviour as the quantity to optimise at every decision point. Our contribution is as much negative as positive. The criterion works, selecting an encoder that in-domain accuracy would have rejected. It is also not enough: a held-out out-of-domain corpus still overstated our blind score by 20 points, and we identify checkpoint selection on a corpus that has entered the training mixture as one mechanism behind that gap.

2 Data

Primary training data. We train on ADI20-micro,[1] the organizer-released subset of ADI-20 (Elleuch et al., 2025a): 67,015 training and 10,806 validation clips over 20 classes.[2] The corpus is balanced by duration rather than clip count, each dialect contributing roughly ten hours, so clip counts vary by up to 2.4× between the largest and smallest class, reflecting differences in mean clip length rather than label imbalance. We use the training split in full and reserve the official validation split for in-domain evaluation. We excluded the full ArabicSpeech/ADI20 release because its size exceeded our compute budget.

Secondary corpora. The shared task provides no development set from the blind test domain, so any estimate of cross-domain behaviour must come from corpora we source ourselves. Casablanca (Talafha et al., 2024) covers 8 of the 20 classes and ships validation and test splits only, with a 1.57× label imbalance. Its `seg_id` field encodes a program identifier, which we treat as a pseudo-source: clips from one program plausibly share a speaker and recording chain, so we split at the program rather than the clip boundary, contributing 10,899 clips to training and holding out the remainder as a program-disjoint evaluation split. Casablanca contains neither Tunisian nor MSA, and Tunisian acts as a false attractor on it.

MADIS-5 (Abdullah et al., 2025b) is a manually curated cross-domain benchmark spanning sources that differ in recording condition. It is labelled over five macro-regions (Egyptian, Gulf, Levantine, Maghrebi, MSA) rather than the 20-way inventory, so we take the `argmax` over the twenty class logits and map the predicted class to its macro-region under a fixed lookup; MSA maps to its own region, so no utterance is discarded and no probability mass is pooled. MADIS-5 accuracy is therefore a five-way figure, not comparable in difficulty to the twenty-way accuracies beside it. We exclude the TV-drama subset, whose acted speech we judged unrepresentative of our target conditions, and evaluate on the radio broadcast, TEDx, and theatre subsets, 3,965 clips in total. Because MADIS-5 is never trained on, it is the only genuinely out-of-domain measurement we hold.

Preprocessing and scope. Audio is decoded and resampled to 16 kHz mono, then passed to the backbone's own `AutoProcessor` feature extractor; we apply no voice-activity detection, loudness normalisation, or transcript filtering. Backbone benchmarking uses a 25% subsample of the training split; the final system uses it in full. Evaluation uses the ADI20-micro validation split, the program-disjoint Casablanca split, MADIS-5, and the official blind test set. Casablanca is out-of-domain during backbone selection but in-corpus for the final system, where its accuracy is no longer evidence of transfer.

3 System

3.1 Backbone Selection

The blind test domain is unknown at development time, so in-domain validation accuracy measures fit to the one domain we know the test set does not come from. We select on out-of-domain accuracy instead.

We compare three backbones under a common architecture: an encoder, mask-aware mean pooling, and a 20-way linear head. The candidates

[1] https://huggingface.co/datasets/UBC-NLP/NADI_2026_ADI20_micro

[2] Strictly, nineteen countries and one MSA class. We follow the shared task in calling the inventory country-level.

Backbone	Steps	ADI20	Casa	MADIS
Whisper-small‡	3.0k	32.87	36.21	18.96
ADI-whisper-ADI20†	3.0k	83.70	83.74	48.02
cohere-transcribe-ar	3.0k	78.12	81.39	80.22
cohere-transcribe-ar	5.2k	90.52	93.99	75.99

Table 1: Backbone comparison, top-1 accuracy (%), identical split and protocol. No selection run trains on Casablanca, so Casablanca is out-of-domain here; it is in-corpus for the final system in Table 2. MADIS-5 is scored over five macro-regions, the others over twenty classes. †Mean of seeds 42/1337, differing by 4.08 points in-domain and 1.20 on MADIS-5. ‡Still improving when stopped, so this is a floor.

are Whisper-small (Radford et al., 2023); ADI-whisper-ADI20 (Elleuch et al., 2025b), a Whisper-large-v3 encoder fine-tuned for ADI and the NADI-2025 winner; and `cohere-transcribe-arabic` (Cassini et al., 2026), a 2B-parameter conformer-based encoder-decoder ASR model for Arabic, of which we use the encoder only. All runs share one split and protocol: 1,381 Casablanca clips and the 3,965 clip MADIS-5 evaluation set described above.

Table 1 shows the two criteria disagreeing. At matched steps ADI-whisper-ADI20 leads on ADI20 by 5.6 points while trailing the Arabic ASR encoder on MADIS-5 by 32. The same tension appears within one backbone: training from 3.0k to 5.2k steps buys 12 points in-domain and costs 4 out-of-domain. We select `cohere-transcribe-arabic` on the out-of-domain criterion, noting that all rows are truncated rather than trained to convergence.

The two criteria also differ in how much of their gap is noise rather than signal. Repeating ADI-whisper-ADI20 across seeds 42 and 1337 moves in-domain accuracy by 4.08 points and MADIS-5 accuracy by 1.20. Against that reference, the 5.6-point in-domain gap between the two leading backbones is barely larger than seed variation, while the 32-point MADIS-5 gap is more than twenty times it. On the evidence available to us the in-domain ranking is not resolvable and the out-of-domain ranking is, which is a second and independent reason to prefer the latter. We have this estimate for one backbone only, so it bounds the comparison rather than settling it.

LoRA does not help: both backbones fall to 15.4% and 16.4% on 20-way Casablanca and the Arabic encoder's advantage vanishes, trailing Whisper-small on MADIS-5 (38.9% vs 43.0%), so its pretrained representation is not dialect-discriminative as it stands and must be adapted by full fine-tuning.[3]

3.2 Training and Final Configuration

The final system is `cohere-transcribe-arabic` with mask-aware mean pooling and a 20-way head, fine-tuned on the full ADI20-micro training split concatenated with the Casablanca training portion. Because Casablanca covers only 8 classes, plain concatenation skews the class distribution, so we sample rows with weight proportional to $\mathrm{count}(y)^{-\alpha}$, $\alpha = 0.5$; we rebalance only partially because $\alpha = 1.0$ would upweight Tunisian and MSA, precisely the two classes absent from the secondary training domain.

The head trains alone for 500 steps while the encoder is frozen, after which the encoder unfreezes with a linear re-warm-up. Head and encoder use learning rates 2×10^{-3} and 2×10^{-5} on cosine decay. We use AdamW ($\beta = (0.9, 0.98)$, $\epsilon = 10^{-6}$), weight decay 0.01 on matrix parameters only, gradient clipping 5.0, label smoothing 0.1, and bf16, with no layer-wise learning-rate decay. Effective batch size is 128 (micro-batch 16, 8 accumulation steps). Each batch is cropped to one duration drawn from {1, 2, 3, 5, 8, 12} seconds, sampled per batch to keep a fixed shape; with blind-test clip lengths unknown, this is our main deliberate robustness measure. We also apply SpecAugment (128 mel bins, two frequency and two time masks) and layer-mixing, a learned weighted sum over encoder layers. Training runs 6,000 steps with evaluation every 400 steps on one NVIDIA RTX 6000, taking roughly four hours per run across eight Arabic-encoder development runs. Checkpoints are selected on Casablanca accuracy, which Section 4.2 shows to be the system's weakest link. At inference each clip is passed at full length without cropping, mean-pooled over unpadded frames, and assigned the `argmax` over twenty logits; no test-time augmentation, chunking, checkpoint averaging, or ensembling is used. Code: `https://github.com/musab-iskandar/Arabic_Spoken_Dialect_Identification`.

Evaluation set	Acc.	C_{avg}
ADI20-micro (in-domain)	91.54	0.046
macro-averaged	92.38	—
Casablanca (in-corpus)	92.97	0.014
scored over 8 classes	95.53	0.031
MADIS-5 (out-of-domain, 5-way)	73.10	—
Blind test (official)	52.51	0.11

Table 2: Final system across evaluation sets, ordered by increasing distance from the training domain. Casablanca is in-corpus, since its training portion is in the training mixture. C_{avg} is computed over the full 20-class output space; the 8-class row restricts scoring to the countries present in Casablanca. MADIS-5 is scored over five macro-regions and is not comparable in difficulty to the 20-way rows.

#	Participant	Acc.	C_{avg}
1	Thakaa	**0.56**	0.11
2	Salesteq	0.55	**0.09**
3	lynx	0.54	0.12
4	**musab-iskandar** (ours)	0.53	0.11
5	oddadmix	0.51	0.13

Table 3: Official leaderboard, top five, as reported (two decimals). Our exact accuracy is 52.51%.

4 Results

4.1 Final System Performance

Table 2 reports the submitted system. The evaluation sets degrade monotonically with distance from the training domain, and that ordering is the result. Casablanca sits slightly above in-domain accuracy rather than below it, which follows once it is recognised as in-corpus. Restricting Casablanca scoring to its eight present classes adds 2.6 points, so part of the residual error there is false positives on absent classes rather than dialect confusion; C_{avg} moves the other way under the same restriction, because those classes contribute only false-alarm terms to the 20-class average and deflate it.

The headline gap is 39.03 points, with C_{avg} rising from 0.046 to 0.11. What matters is how little of it any development signal anticipated. Moving from the in-domain validation split to the held-out MADIS-5 corpus costs 18.44 points, and the further move to the blind test set costs 20.59 more. These two figures do not decompose the gap additively, because MADIS-5 is scored over five macro-regions while the other two sets are scored over twenty classes, and a five-way score flatters the model. A twenty-way MADIS-5 score would be lower, so 18.44 is a lower bound on the domain-shift penalty as measured on MADIS-5, and 20.59 an upper bound on the residual that no development signal exposed. Either way, our best out-of-domain estimate was optimistic by roughly twenty points.

Table 3 puts that gap in context. The four leading systems span three points of accuracy and the top five span five, against our own 39-point in-domain-to-blind gap, so the task is dominated by domain transfer rather than by differences between the systems entered into it. The two metrics do not even agree on the ordering: ranked by C_{avg}, the second-placed system leads at 0.09 and our 0.11 ties the winner, while the third-placed system falls behind both. The two respond differently to false alarms on classes absent from the blind subset, a measurable share of our errors (Section 4.3).

4.2 Checkpoint Selection

Across the eight Arabic-encoder development runs, out-of-domain accuracy peaks very early and then declines while in-domain accuracy keeps climbing. MADIS-5 peaks at step 400 in six of eight runs and by step 800 in the other two, reaching 79.9–86.0%, then falls by a mean of 4.78 points (max 8.3) by the step each run actually selected. In-domain and Casablanca accuracy rise monotonically over the same interval. That 4.78 points is the measurable cost of the selection rule: what an oracle stopping on MADIS-5 would have recovered for free. Selecting on Casablanca accuracy steers away from the best out-of-domain checkpoint because Casablanca is in-corpus for the final system, so its accuracy tracks in-domain fit rather than transfer and the selection rule inherits that bias.

4.3 Error Analysis

The closed label space is a measurable source of error independent of dialect confusion. On Casablanca, which contains 8 of the 20 classes, restricting scoring to those 8 raises accuracy from 92.97% to 95.53%, so 2.56% of clips are assigned to classes that cannot occur in that corpus, with Tunisian the dominant false attractor: it is absent from Casablanca yet remains a high-prior class in the training mixture. Since the blind set also covers an undisclosed subset of the inventory, a comparable share of blind errors is likely to be false positives on absent classes rather than confusion between plausible neighbours. This also bounds what class-rebalancing can buy, and is why we set $\alpha = 0.5$ rather than equalising priors: the classes

[3] LoRA runs used a different split (1,365 Casablanca, 1,997 MADIS-5 clips) and are not directly comparable to Table 1.

full balancing would upweight are precisely those most likely to fire spuriously off-domain.

5 Discussion

5.1 Why the development estimate failed

Casablanca was out-of-domain during backbone selection and gave a clean signal there, correctly preferring an encoder that in-domain accuracy would have rejected. Once its training portion entered the final mixture, the same corpus became a measure of in-domain fit, and the selection rule silently changed meaning without changing its name. A corpus serves as a model-selection signal only for as long as it stays outside training, and mixing it in for the accuracy gain costs the diagnostic that justified choosing the model in the first place.

We did not submit a contrastive run that excludes Casablanca from training, but the selection runs in Table 1 are exactly that configuration, and the comparison points in an uncomfortable direction. The Arabic encoder trained on ADI20-micro alone reaches 75.99 on MADIS-5; the final system, which adds the Casablanca training portion, reaches 73.10. This is not a controlled ablation, since the selection run uses a 25% subsample and stops at 5.2k steps while the final system uses the full split and runs to 6k. Read cautiously, though, the Casablanca mixture bought in-corpus accuracy and did not buy out-of-domain accuracy, which is consistent with the selection failure above rather than in tension with it.

5.2 Next steps

Three changes, in order of expected value. First, reserve a corpus that is never trained on for checkpoint selection, and run the ADI20-only ablation above so that the cost of the Casablanca mixture is measured rather than inferred. Second, investigate the early out-of-domain peak: if it is a genuine optimum rather than an early-training artefact, early stopping on a held-out out-of-domain signal recovers several points at no cost, and Section 4.2 puts that figure at 4.78 points on average for our runs. Third, attack the nuisance factors our system does nothing to suppress. The literature converges here. Domain-adversarial training through a gradient reversal layer (Ganin et al., 2016; Sun et al., 2018) improves cross-corpora error in LID, and Dey et al. (2023) report that it is specifically the domain-invariant formulation that transfers, the domain-aware variant improving same-corpus performance instead. Channel-invariance objectives achieve a similar effect without requiring domain labels (Muralikrishna and Dinesh, 2022). Speaker-normalising voice conversion (Abdullah et al., 2025b) raised cross-domain accuracy substantially in NADI-2025 (Abdullah et al., 2025a) and targets speaker identity directly. Repairing the augmentation path is a precondition for testing any of these, since a silent no-op would make each of them look ineffective for the same reason it made augmentation look neutral.

6 Conclusion

We submitted an Arabic ASR encoder with mean pooling and a 20-way linear head, trained on ADI20-micro mixed with Casablanca and selected on out-of-domain rather than in-domain accuracy, placing fourth on NADI-2026 Subtask 2 with 52.51% accuracy and a C_{avg} of 0.11. The criterion worked where we applied it consistently, choosing an encoder that in-domain accuracy would have rejected by 32 points, more than twenty times our measured seed variation. It was not sufficient: our best development estimate still overstated the blind score by 20 points, and at least 18.44 of the 39.03-point gap is domain shift. Two things follow. A corpus stops being a valid selection signal the moment it enters the training mixture, however tempting the accuracy it buys. And out-of-domain accuracy peaks within the first few hundred steps and then decays while in-domain accuracy climbs, so the best-generalising checkpoint is rarely the one an in-domain criterion finds.

Limitations

Our conclusions rest on one submitted system and one blind score, and each backbone in Table 1 is a single truncated run, so the ranking could reorder with longer training. The seed comparison in Section 3.1 bounds run-to-run variation for one backbone only; we did not repeat the others.

MADIS-5 is scored over five macro-regions rather than the official twenty-class inventory, so it is not directly comparable to the other evaluation sets. The two figures in Section 4.2 and Section 3.1 are consequently bounds and not a decomposition: 18.44 is a lower bound on the domain-shift penalty as measured on MADIS-5, and 20.59 an upper bound on the unexplained residual. A twenty-way out-of-domain corpus would settle this, and we did not have one.

Casablanca serves two roles that are in tension. It is out-of-domain during backbone selection, where its signal is valid, and in-corpus for the final system, where it is not. We report both roles explicitly, but the consequence is that the checkpoint-selection rule for the submitted system rested on a corpus that had already entered training. Separately, the backbone's release benchmarks include Casablanca, so we cannot rule out pretraining exposure rather than transfer as an explanation for its performance there.

Compute was the binding constraint on scope. All development ran on a single NVIDIA RTX 6000, roughly four hours per run across eight Arabic-encoder runs, and two choices follow from that budget rather than from any experimental finding. We trained on ADI20-micro rather than the full ArabicSpeech/ADI20 release, so we cannot say how much of the gap to the blind set is a data-scale effect. And we submitted a single model: ensembling across backbones or architectures would require training each candidate to convergence rather than truncating it, which we could not afford, and the disagreement between backbones in Table 1 suggests their errors are far from identical. Both are plausible sources of the remaining headroom.

Finally, our configuration requested four waveform augmentations, but a post-hoc audit found the dispatch table was never constructed: every call raised a `NameError` that a broad exception handler absorbed, returning the unmodified waveform. All results here are therefore an un-augmented baseline, and asserting that an augmented waveform differs from its input would have caught it.

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